

t-SNE: t-distributed Stochastic Neighbor Embeddings

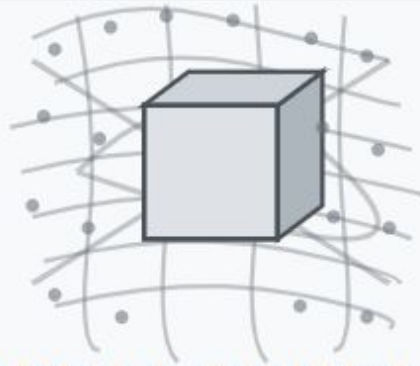
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CSE437

We live in a world full of data.

But how can we “see” high-dimensional data — something that has hundreds or thousands of features?



1000 FEATURES



"Where do I even start?"



Gene Expression

cat [0.2, -0.5, 0.8, ...]

dog [0.2, -0.5, 0.8, ...]

car [0.2, -0.5, 0.8, ...]

tree [0.2, -0.5, 0.8, ...]

Word Embeddings

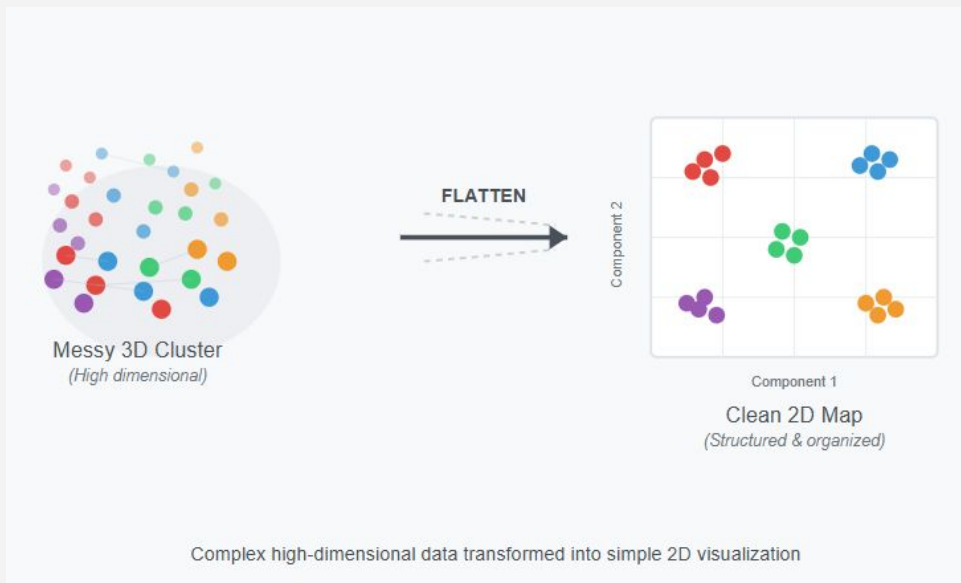


Images

100+ dimensions

Why We Need Dimensionality Reduction

Dimensionality reduction helps us see complex data in simpler form. It's like summarizing a movie into a 2-minute trailer — we keep the essence, not all the details.



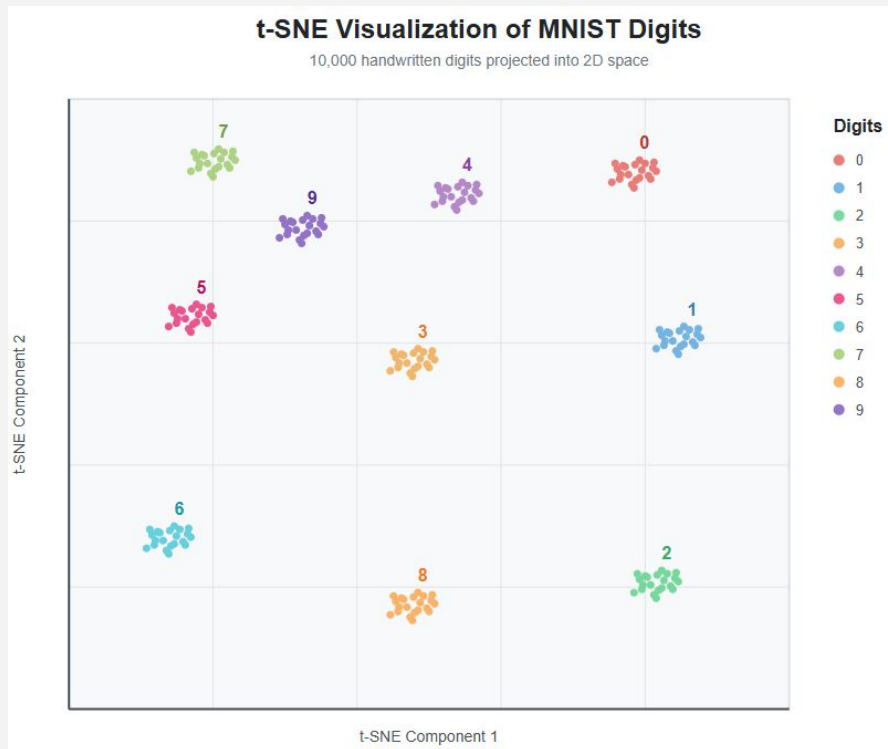
Meet t-SNE

t-SNE (t-Distributed Stochastic Neighbor Embedding) is a smart algorithm that:

Turns high-dimensional data into 2D or 3D

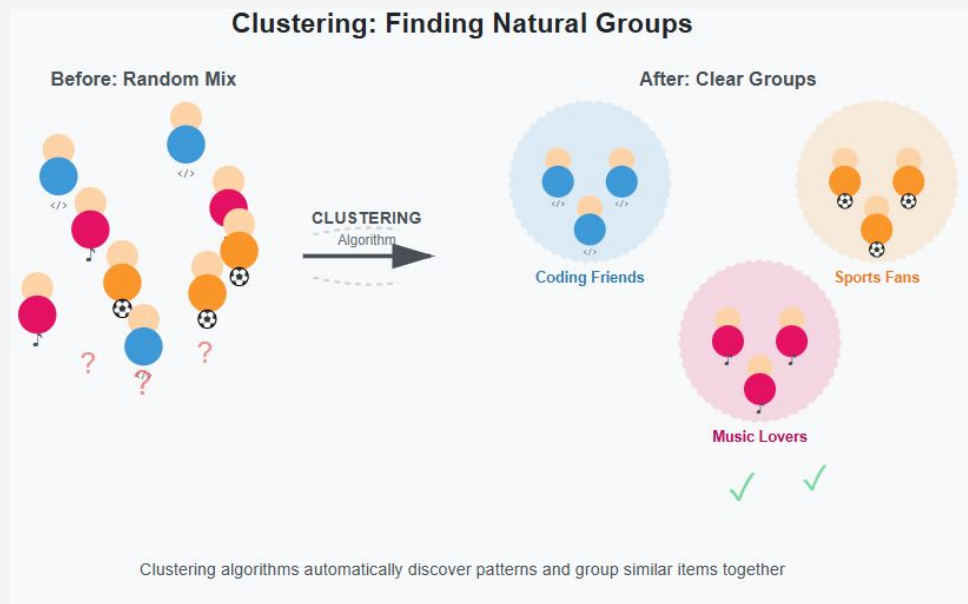
Keeps similar items close

Pushes different items apart



Let's Imagine a Story

Imagine 1000 students — each with different hobbies, skills, and interests.
Every student is a point in a 1000-dimensional space!
t-SNE helps find groups: gamers, artists, coders, musicians...



The Magic of t-SNE (Simple Explanation)

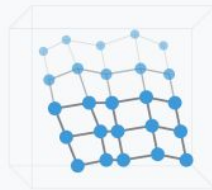
t-SNE looks at who is close to whom in high dimensions.

Then it tries to draw the same friendships in 2D.

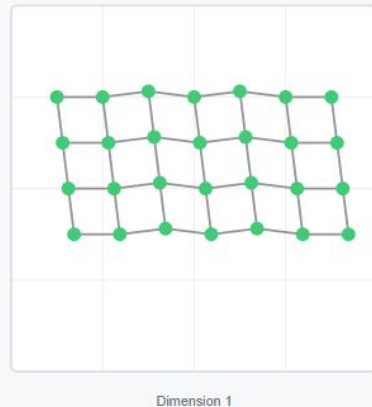
- Close friends = stay close
- Strangers = stay apart

Manifold Learning: Preserving Neighborhood Structure

High-Dimensional Space
(shown as 3D projection)



2D Map
(connections preserved)

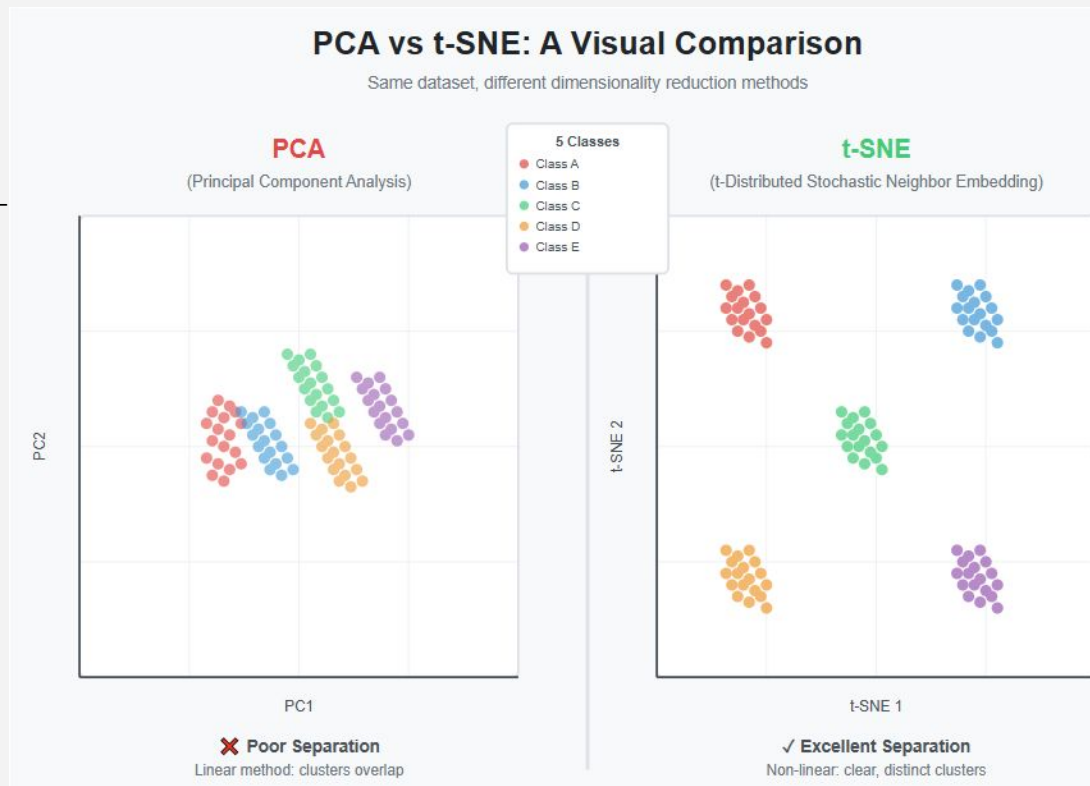


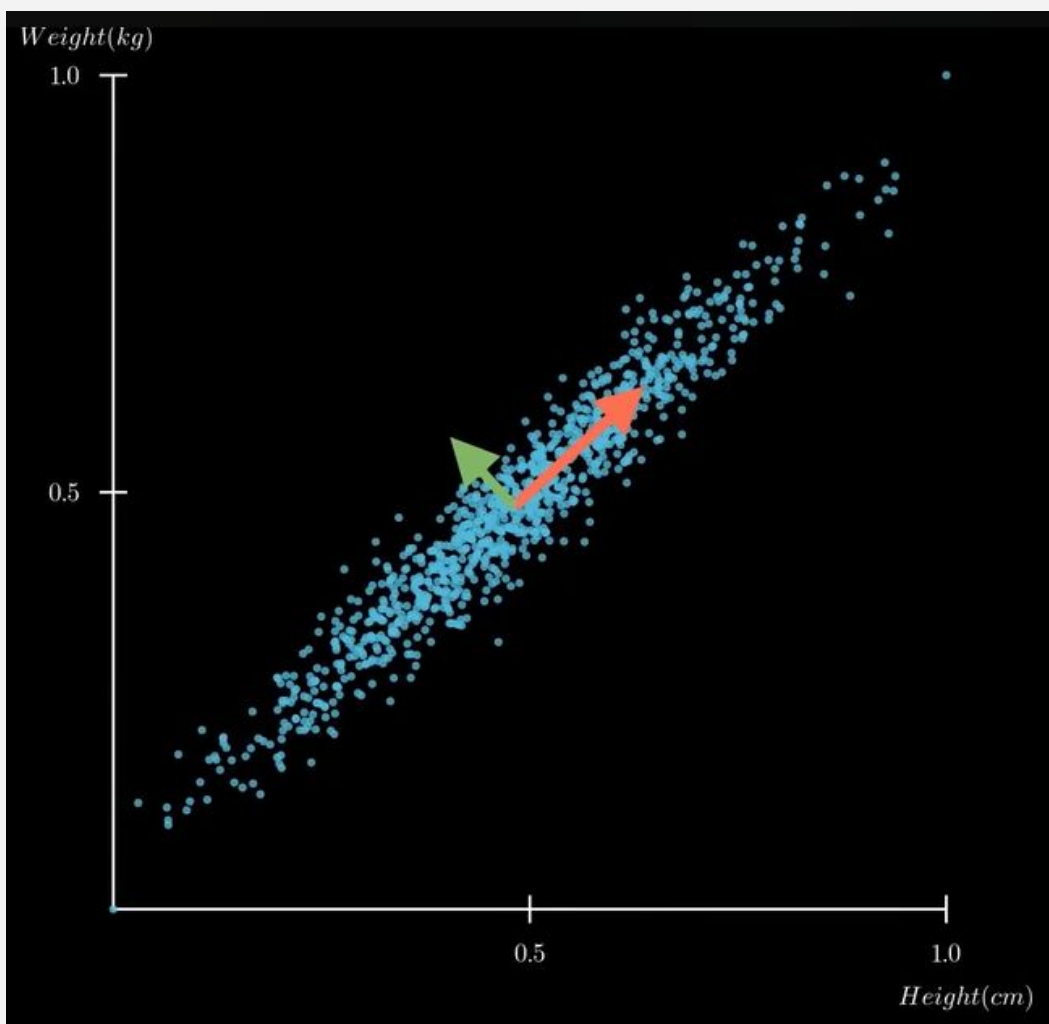
PROJECTION
Keep neighbors close!

Connected neighbors in high dimensions remain connected neighbors in 2D

Why Not Just Use PCA?

PCA is like flattening a 3D object into 2D — simple, but loses shape.
t-SNE is like sculpting — it keeps the small local details intact.





Breaking Down the Name

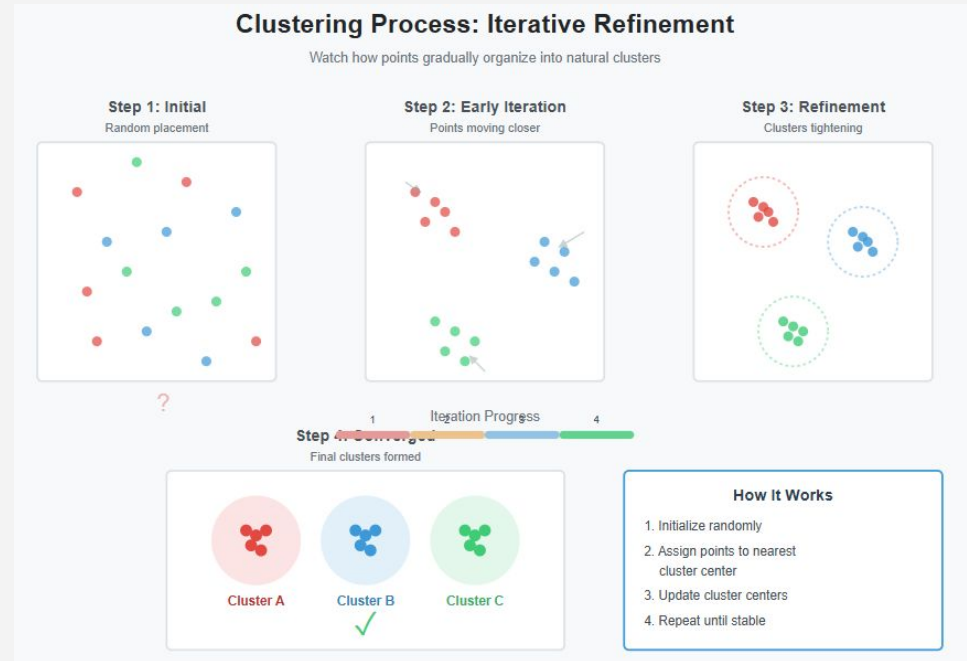
Word	Meaning
t-Distributed	Uses a special formula to spread points apart and reduce crowding
Stochastic	Adds randomness to explore patterns better
Neighbor	Focuses on local “friends” or nearby points
Embedding	Maps data into a smaller, visible world

What t-SNE Actually Does (Intuition)

Measures how close points are in high dimensions

Puts points randomly in 2D

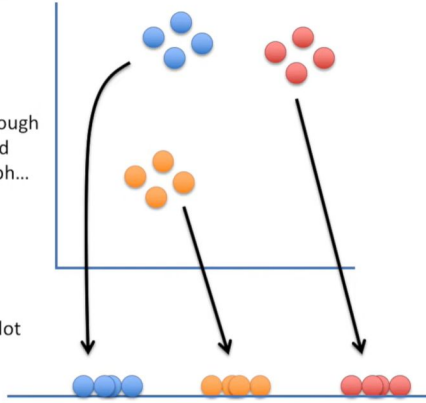
Moves them little by little until neighborhoods are preserved



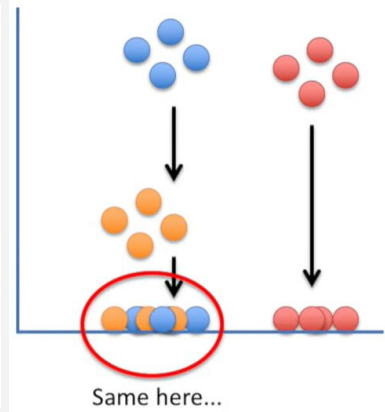
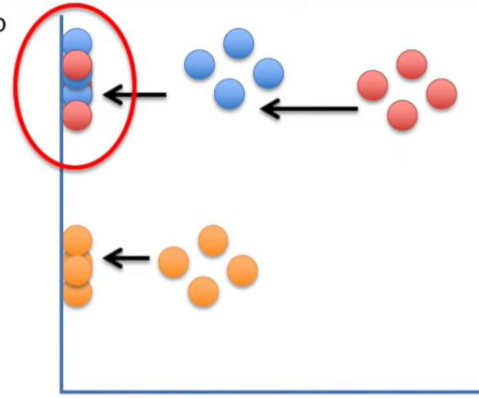
Here's a basic 2-D scatter plot.

Let's do a walk through of how t-SNE would transform this graph...

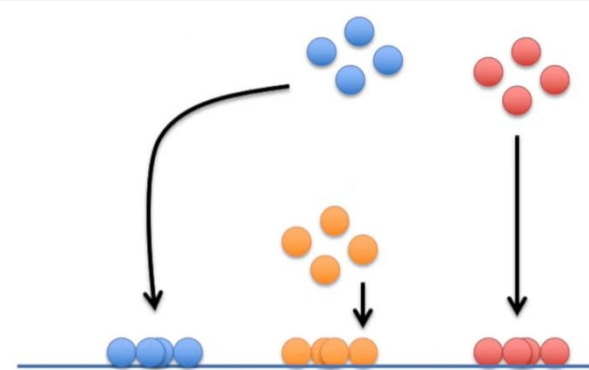
...into a flat, 1-D plot on a number line.



Instead of two distinct clusters, we just see a mishmash.



From here on out, t-SNE moves these points, a little bit at a time, until it has clustered them.



What t-SNE does is find a way to project data into a low dimensional space (in this case, the 1-D number line) so that the clustering in the high dimensional space (in this case, the 2-D scatter plot) is preserved.

Non-linear Capabilities

High-D: Curved manifold



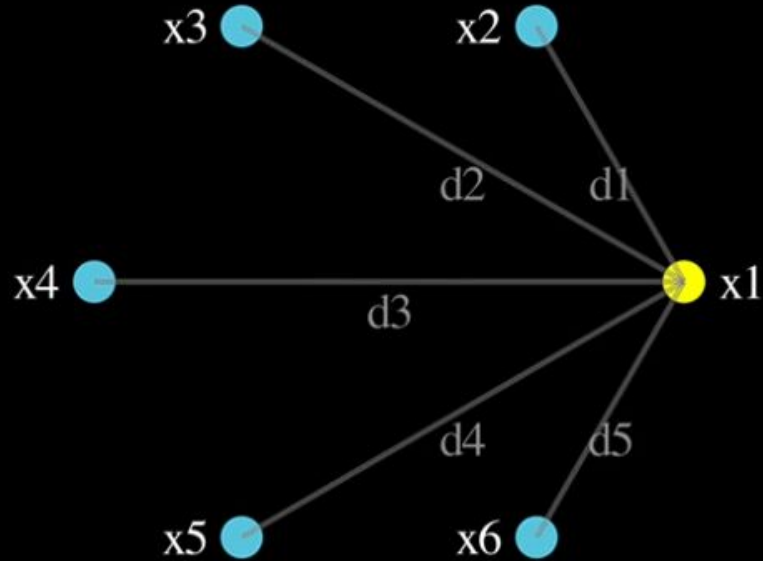
t-SNE



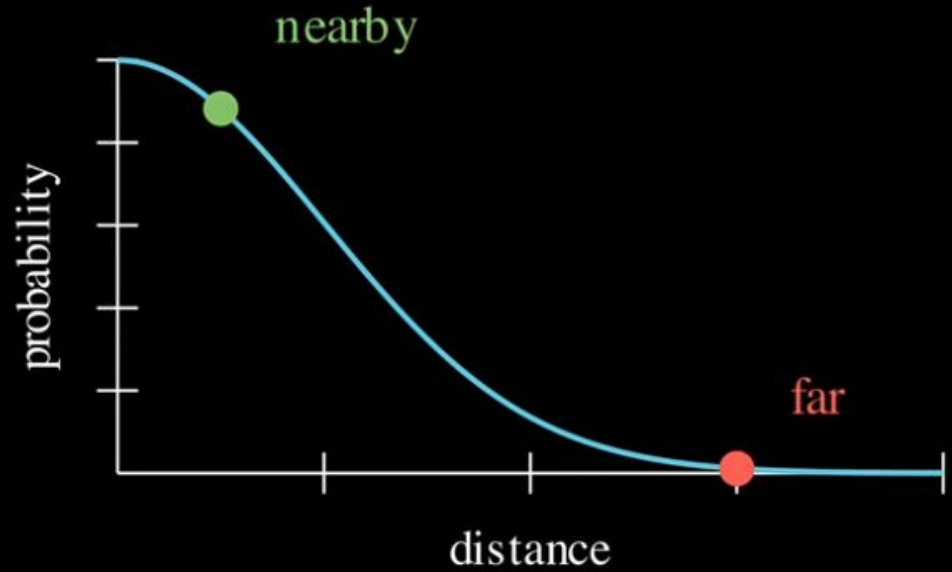
2D: Unfolded



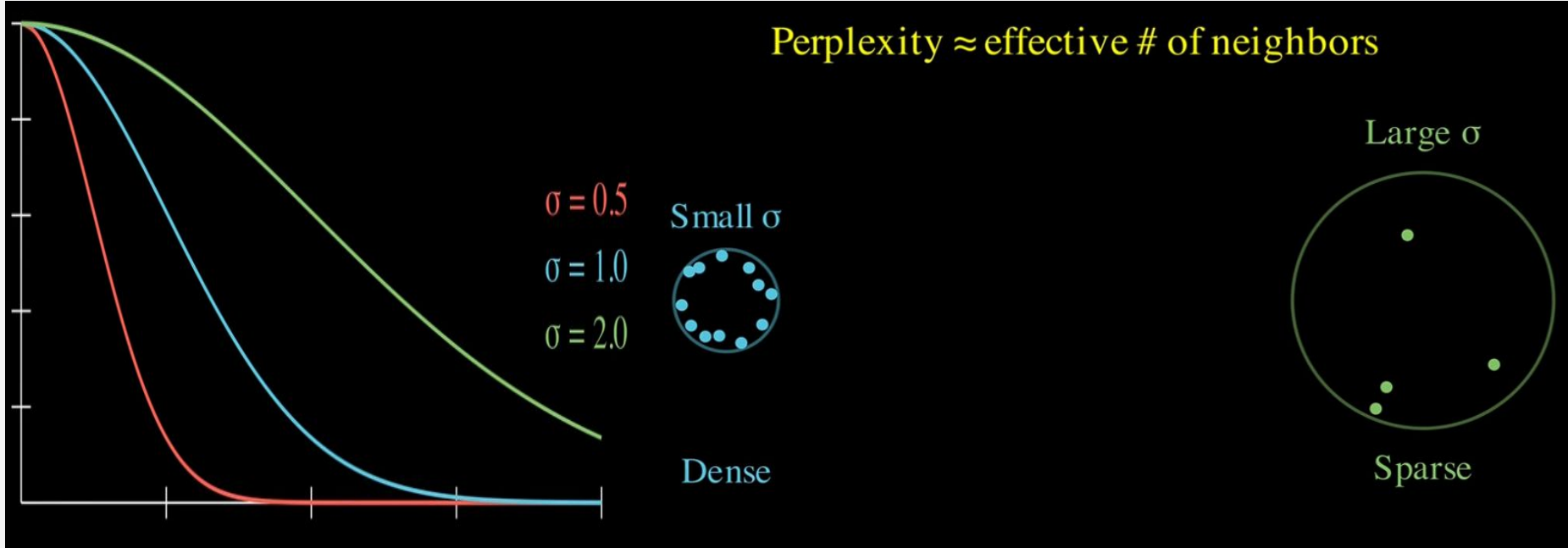
Linear methods miss this structure



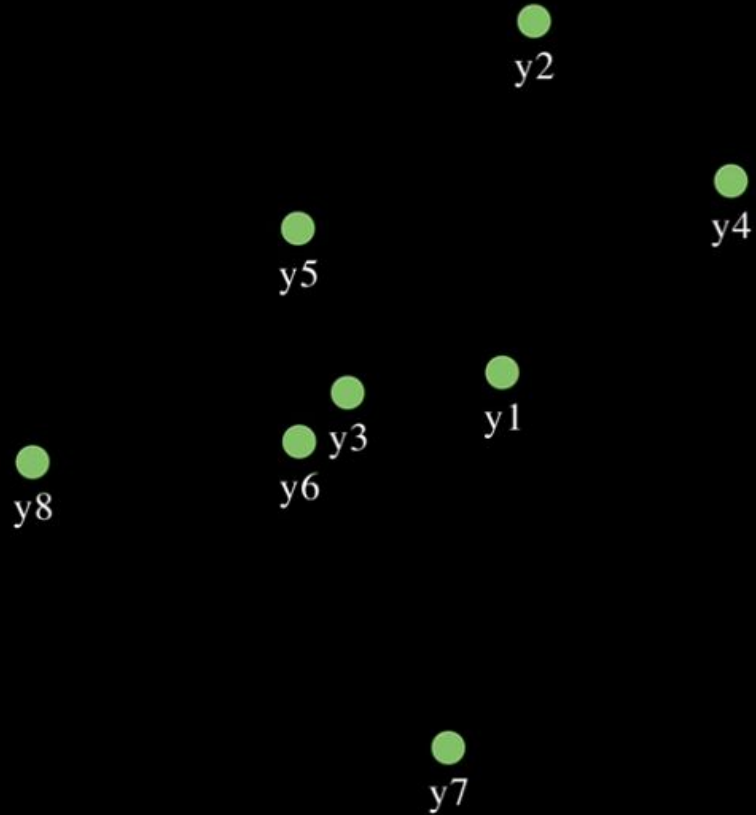
$$p_{j|i} = \frac{e^{-d_{ij}^2/2\sigma^2}}{\sum_k e^{-d_{ik}^2/2\sigma^2}}$$



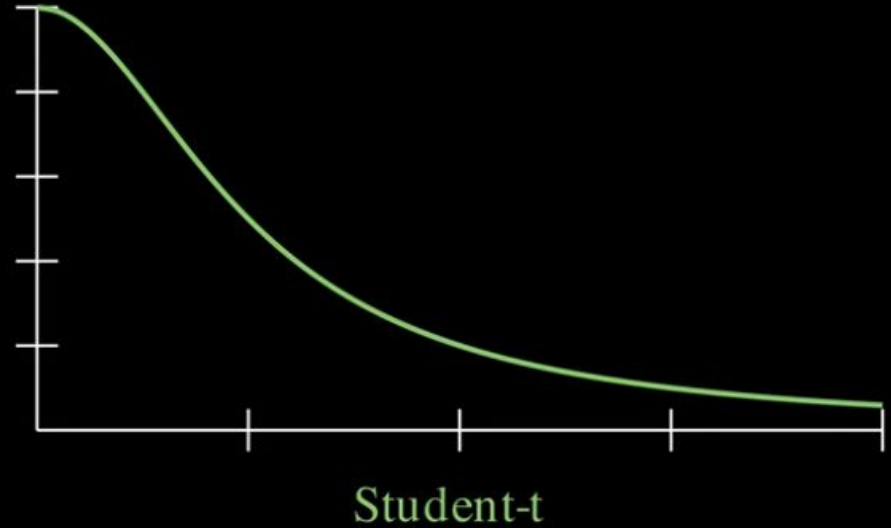
Perplexity $\approx$ effective # of neighbors

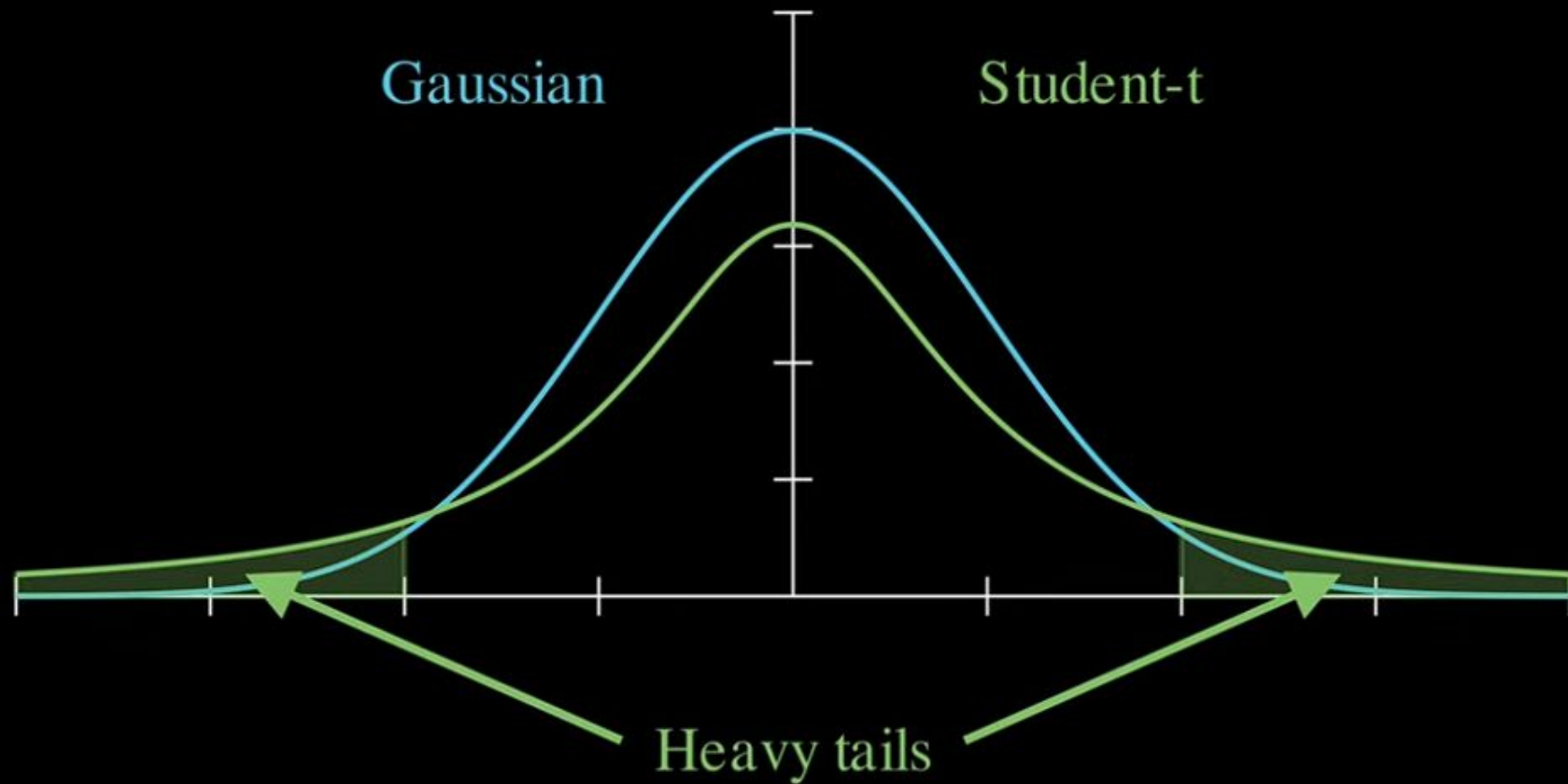


Random initialization

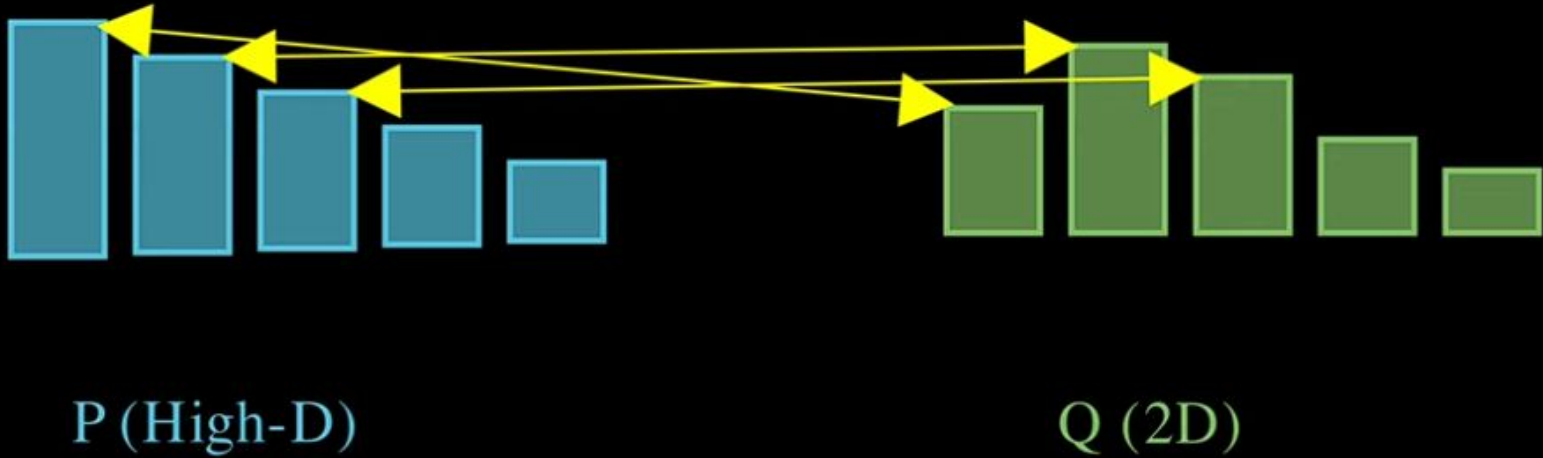


$$q_{ij} = \frac{(1 + d_{ij}^2)^{-1}}{\sum_{kl} (1 + d_{kl}^2)^{-1}}$$





$$KL(P||Q) = \sum_{ij} p_{ij} \log \frac{p_{ij}}{q_{ij}}$$



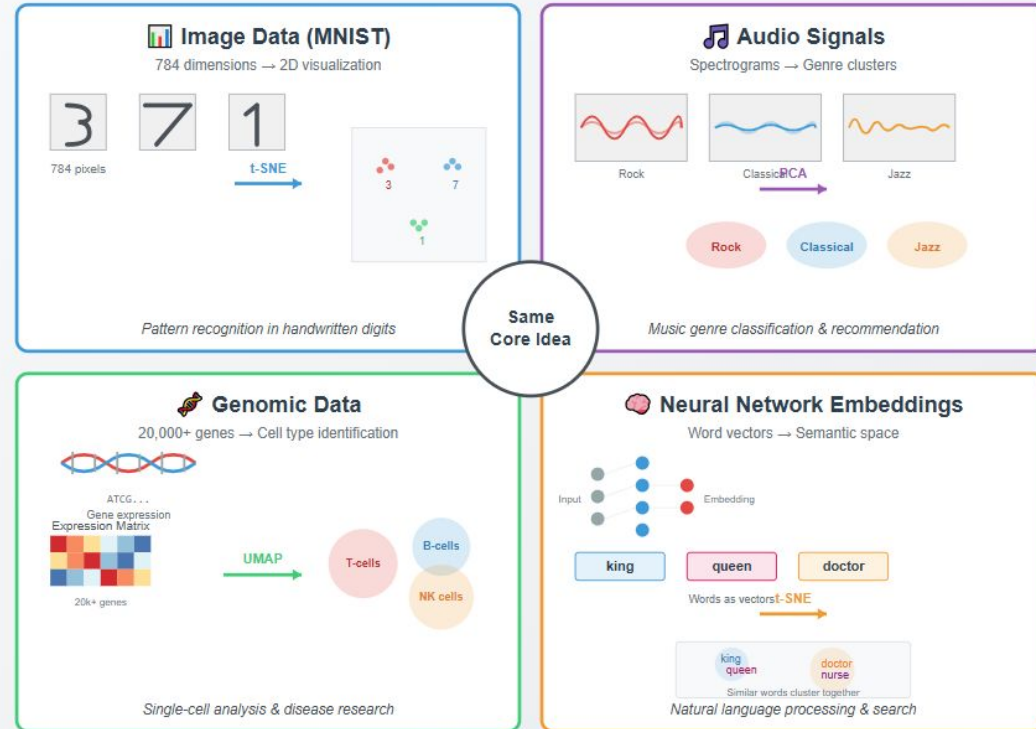
Measures difference

Real-World Uses of t-SNE

- Visualizing handwritten digits (MNIST)
- Finding similar songs or artists
- Exploring gene expression in biology
- Understanding how neural networks learn

Applications of Dimensionality Reduction

From pixels to proteins: making sense of high-dimensional data



t-SNE in Action

<https://distill.pub/2016/misread-tsne/>

Key Takeaways

- ✓ t-SNE helps us see invisible patterns in data
- ✓ It's great for visualization, not prediction
- ✓ It focuses on local relationships
- ✓ It's more powerful than PCA for complex data

Thank You!

HAVE A GREAT BREAK